

Step 1: Import Required Libraries

```
import numpy as np
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

Step 2: Create the Dataset

```
data = {
    'pH': [7.2, 6.8, 5.5, 8.8, 7.0],
    'Turbidity': [1.5, 2.0, 8.2, 6.5, 1.0],
    'Dissolved_Oxygen': [8.5, 7.8, 3.5, 4.0, 9.0],
    'Target': [1, 1, 0, 0, 1]
}

df = pd.DataFrame(data)

print(df)
```

	pH	Turbidity	Dissolved_Oxygen	Target
0	7.2	1.5	8.5	1
1	6.8	2.0	7.8	1
2	5.5	8.2	3.5	0
3	8.8	6.5	4.0	0
4	7.0	1.0	9.0	1

Step 3: Separate Features and Labels

```
X = df[['pH', 'Turbidity', 'Dissolved_Oxygen']]

y = df['Target']
```

Step 4: Split Training and Testing Data

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

Step 5: Feature Scaling

```
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)
```

Step 6: Train the Single-Layer Perceptron

```
model = Perceptron(
    max_iter=1000,
    eta0=0.1,
```

```

        random_state=42
    )

    model.fit(X_train,y_train)

```

▼ Perceptron ⓘ ?

```
Perceptron(eta0=0.1, random_state=42)
```

Step 7: Display Learned Weights

```

print("Weights")

print(model.coef_)

print()

print("Bias")

print(model.intercept_)

```

```

Weights
[[ 0.13899173 -0.12525374  0.10945409]]

Bias
[-0.1]

```

Step 8: Make Predictions

```

y_pred = model.predict(X_test)

print("Actual :", y_test.values)

print("Predicted :", y_pred)

```

```

Actual : [1]
Predicted : [1]

```

Step 9: Evaluate the Model

```

accuracy = accuracy_score(y_test,y_pred)

print("Accuracy :",accuracy)

print()

print("Confusion Matrix")

print(confusion_matrix(y_test,y_pred))

print()

print(classification_report(y_test,y_pred))

```

```
Accuracy : 1.0
```

```
Confusion Matrix
[[1]]
```

	precision	recall	f1-score	support
1	1.00	1.00	1.00	1
accuracy			1.00	1
macro avg	1.00	1.00	1.00	1

```
weighted avg      1.00      1.00      1.00      1
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:407: UserWarning: A single label was  
warnings.warn(
```

Step 10: Predict a New Water Sample

```
new_sample = np.array([[7,2,8]])  
  
new_sample = scaler.transform(new_sample)  
  
prediction = model.predict(new_sample)  
  
if prediction[0] == 1:  
    print("Safe Water")  
else:  
    print("Unsafe Water")
```

Safe Water

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid fe  
warnings.warn(
```

Step 11: Display the Final Dataset

```
df['Prediction'] = model.predict(scaler.transform(X))  
  
print(df)
```

	pH	Turbidity	Dissolved_Oxygen	Target	Prediction
0	7.2	1.5	8.5	1	1
1	6.8	2.0	7.8	1	1
2	5.5	8.2	3.5	0	0
3	8.8	6.5	4.0	0	0
4	7.0	1.0	9.0	1	1